

Demonstration of Network Communication Models

OSI & TCP/IP Protocol Analysis using Cisco Packet Tracer

1. Aim & Objectives

Aim: To analyze and demonstrate how data travels through the OSI model and TCP/IP model layers using the Cisco Packet Tracer simulation environment.

Key Objectives:

- Understand the structure and functionality of the 7-layer OSI model and the 4-layer TCP/IP model.
- Design and deploy a functional local area network (LAN) topology in Cisco Packet Tracer.
- Observe and analyze real-time packet movement, frame construction, and data encapsulation/decapsulation across layers.
- Verify successful end-to-end communication between network devices using diagnostic commands.

2. Selected Network Scenario & Topology

The case study is evaluated using a standard star-topology LAN configuration:

Hardware & Devices

- **Switch:** Cisco Catalyst 2960-24TT (1 Unit)
- **End Devices:** Standard Workstation PCs (2 Units)
- **Cables:** Copper Straight-Through

IP Addressing Scheme

- **PC0:** 192.168.1.10 / 255.255.255.0
- **PC1:** 192.168.1.20 / 255.255.255.0
- **Port Mapping:** PC0 → Fa0/1 | PC1 → Fa0/2

3. Layered Communication Models Comparison

Layer	OSI Layer Name	Functional Description	TCP/IP Layer	Protocols
7	Application	User application interaction & network services	Application	HTTP, DNS, FTP
6	Presentation	Data formatting, encryption, and compression		SSL, TLS, JPEG
5	Session	Establishes, manages, and terminates sessions		NetBIOS, RPC
4	Transport	End-to-end delivery, segmentation, error recovery	Transport	TCP, UDP
3	Network	Logical IP addressing and packet routing	Internet	IP, ICMP, ARP
2	Data Link	Physical MAC addressing and framing	Network Access	Ethernet, Wi-Fi
1	Physical	Electrical signals, bitstream transmission		RJ-45, Cables

4. Experimental Setup & Simulation Procedure

- Topology Assembly:** Place two PCs (PC0, PC1) and one Cisco 2960 Switch in the workspace. Connect FastEthernet interfaces using Copper Straight-Through cables.
- IP Configuration:** Assign static IPv4 addresses (PC0: 192.168.1.10, PC1: 192.168.1.20) with subnet mask 255.255.255.0 via Desktop → IP Configuration.
- Connectivity Verification:** Open PC0 Command Prompt and execute `ping 192.168.1.20`. Verify ICMP echo reply responses.
- Simulation Activation:** Switch mode from *Realtime* to *Simulation Mode*.
- PDU Tracking:** Select 'Add Simple PDU', set PC0 as source and PC1 as destination. Use 'Capture/Forward' to observe frame traversal step-by-step.

5. Analysis of Data Encapsulation Process

When sending a message (e.g., ICMP Request or "Hello" payload) from PC0 to PC1, data flows downwards through layers at sender side and upwards at receiver side:

- Application** → User payload message ("Hello") is generated.
- Transport** → Data is segmented and header added (Source/Destination Ports).
- Network** → Packets encapsulated with IP Header (Source IP: 192.168.1.10, Dest IP: 192.168.1.20).
- Data Link** → Framed with MAC addresses (Source MAC & Destination MAC).
- Physical** → Converted to physical electrical signals and transmitted over cable.

**At PC1, decapsulation strips headers layer-by-layer to deliver original data to the receiving application.*

6. Results & Conclusion

Results: The ICMP packet transmission and simulation inspection verified successful layer-by-layer encapsulation. Ping tests confirmed 0% packet loss, demonstrating proper MAC table learning by Switch0 and correct IP addressing on end systems.

Conclusion: The experiment effectively illustrated the practical operations of OSI and TCP/IP networking models. While the 7-layer OSI model serves as an indispensable theoretical reference framework, the 4-layer TCP/IP suite provides the actual protocol implementations used in modern computer networks.